Discrete Drainage Dynamics II: Cascade-Throttled Drainage and the Emergence of the Schwarzschild Time-Dilation Factor

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April 2026

Abstract

We extend the local rule of Paper I by a single physical modification: the directional flux on a link is throttled by the bandwidth available at the source node. A node with reduced reserve R pushes flux at a rate proportional to R/R_0; the receiver imposes no further limitation, since accepting flux is passive. Under this modification, the steady-state field around a point sink in a substrate at rest satisfies the algebraic identity $\nabla^2(R^2) = 0$ in the continuum limit, away from the sink. The unique spherically-symmetric solution with $R \to R_0$ at infinity is $R/R_0 = \sqrt{1-\beta/r}$, with β fixed by the source strength. This is precisely the form of the Schwarzschild time-dilation factor, $\sqrt{1-2GM/c^2r}$, under the identification $\beta = 2GM/c^2$. Combined with the operational identification of effective proper time with local bandwidth ($d\tau/dt = R/R_0$), the model reproduces the gravitational time dilation of weak-field general relativity not as a leading-order approximation, but as a strict consequence of the bandwidth-throttled cascade. We verify the result numerically. We do not derive Newton's G, the speed of light c, or the full Einstein equations; we derive the static spherically symmetric time-dilation factor, in the continuum limit, modulo the choice of throttle prescription, which we discuss.

1 What this paper claims

1. **Bandwidth identification.** Within the rule of Paper I, the maximum directional flux a node i can push out is αR_i, set by the locally available reserve. We identify the dimensionless ratio R_i/R_0 with the local rate of effective proper time:

$$\frac{d\tau_i}{dt} = \frac{R_i}{R_0}.$$

This identification is operational, not derived; it states that a node's local rate of activity per coordinate tick is proportional to its available reserve.

2. **Single-throttle modification.** We extend the local rule of Paper I by throttling each directional flux on the source side by the source's bandwidth fraction:

$$F_{i\to j} = \alpha \frac{R_i}{R_0} (R_i - R_j)_+.$$

The receiver imposes no additional throttle: pushing flux costs the source bandwidth, accepting flux is passive at the receiver. We adopt this prescription as physically motivated but not uniquely derived; we discuss alternatives in Sec. 4.2.

3. **Schwarzschild form, derived.** In the continuum limit, the steady-state field of the throttled rule around a point sink satisfies $\nabla^2(R^2) = 0$ away from the sink. The unique spherically symmetric solution with $R \to R_0$ at infinity is

$$\frac{R(r)}{R_0} = \sqrt{1 - \frac{\beta}{r}}, \qquad \beta = \frac{\kappa E_0}{2\pi\,\alpha\,R_0}.$$

Combined with the bandwidth identification, this gives the gravitational time-dilation factor

$$\frac{d\tau(r)}{dt} = \sqrt{1 - \frac{\beta}{r}},$$

which under the calibration $\beta = 2GM_{\text{phys}}/c^2$ matches the weak-field Schwarzschild factor at all orders, not just the leading order. The match is exact in the continuum limit, modulo the throttle prescription.

What this paper does not claim.

We do not derive special relativity, and we do not address kinematic (motion-induced) clock-rate effects; those are deferred to Paper III. We do not derive the full Schwarzschild metric; we derive its time-time component for a static spherically symmetric source in the weak-field limit. We do not derive Newton's gravitational constant G or the speed of light c; they enter as calibrations, exactly as in Paper I. We do not show that the rule is unique. We do not derive the single-throttle prescription itself: it is a physically motivated choice, justified by the source-receiver asymmetry of flux exchange, but the rule of Paper I as written does not select it uniquely. Different throttle prescriptions yield different steady-state forms, as illustrated in Sec. 6; the Schwarzschild form is the one selected by the single-throttle choice.

2 The picture, in plain words

Picture each node as a tank holding a non-negative amount of stuff at level R, with a rest level R_0. The pipes between neighbours carry flow whose magnitude depends on the level difference. There is one limit on what any tank can deliver: you cannot push out more stuff than the tank actually holds. A full tank can sustain the rated flow αR_0. A half-full tank can sustain at most $\alpha R_0/2$. An empty tank can deliver nothing. The rate at which a tank can engage with its environment is bounded by the level it currently holds, and not by anything else.

Now think about local time at the tank. The rule provides no hidden mechanism, no internal oscillator, no auxiliary phase. A tank's participation in the rule is entirely its flux activity. We take the rate of effective proper time at a tank to be the rate at which it can support that activity per coordinate tick. Full tank: rated rate. Half-full: half rate. Empty: no rate. Proper time *is* the throughput.

Drop a sink near a region of full tanks. The sink absorbs. Neighbouring tanks lose stuff to the sink. Their levels drop. Their throughput capacity drops. Their effective proper time slows down. The neighbours of the neighbours feel the same pull, and the cascade propagates outward through the lattice.

Now ask the question that gives the paper its central result. The cascade itself is throttled by bandwidth: a tank that has been drained can push less stuff than a full tank, even for the same level difference. This is the physically natural form of the rule — the source needs reserve to push, the receiver does not need anything to accept. So the flux on a link is not just proportional to $(R_i - R_j)$, it is proportional to $(R_i/R_0)(R_i - R_j)$: the source's bandwidth limits its own emission rate.

What does the cascade settle into? Far from the sink, levels are barely perturbed and the throttle is invisible. The leading behaviour is the $1/r$ profile of Paper I. But near the sink, the throttle bites: the cascade has to work harder to maintain the same flux through a region of

reduced bandwidth. The result is a profile that is, in the continuum limit, exactly $\sqrt{1-\beta/r}$, the Schwarzschild form. The body of the paper makes this precise.

3 Setup

We work in the framework of Paper I. The substrate is a graph with at least six neighbours per node and statistical isotropy at large scale. Each node i carries a non-negative reserve $R_i \geq 0$ with rest value R_0. Each oriented link (i,j) carries a non-negative directional flux $F_{i\to j} \geq 0$. The rule alternates a Link Update and a Node Update with a per-node rate-limiter $\beta_i \in [0,1]$ ensuring $R_i \geq 0$ at all times.

Definition (coordinate time). Coordinate time t is the tick count of the substrate.

Definition (bandwidth). The local bandwidth of node i is the dimensionless ratio

$$\mathcal{B}_i \;=\; \frac{R_i}{R_0}. \tag{1}$$

At rest $\mathcal{B}_i = 1$. In the static gravitational regime considered below, $R_i \leq R_0$ everywhere because the only source acts as a sink, so $\mathcal{B}_i \in [0,1]$.

Hypothesis (bandwidth identification). We adopt the operational definition

$$\frac{d\tau_i}{dt} \;=\; \mathcal{B}_i \;=\; \frac{R_i}{R_0} \tag{2}$$

for the local rate of effective proper time. The identification is structural in the sense that it does not introduce a new variable beyond the reserve; it is also assumed in the sense that we do not derive from a deeper principle that "effective proper time" must mean "available throughput". We proceed under this assumption and discuss what is and is not derived in the body of the paper.

Probe approximation. Throughout this paper we treat the test object as a passive probe whose own contribution to the reserve field is negligible compared to the externally generated deficit. The full self-consistent treatment is left to Paper IV.

4 The bandwidth-throttled rule

4.1 Source-side throttle: physical motivation

The flux on a link in Paper I is the textbook discrete diffusion form, $F_{i\to j} = \alpha(R_i - R_j)_+$, proportional to the level difference. We modify it by throttling the source side:

$$F_{i\to j} \;=\; \alpha\,\frac{R_i}{R_0}\,(R_i - R_j)_+. \tag{3}$$

The motivation is direct. Pushing reserve out of node i through a pipe takes resources at i. A drained node has fewer resources; it pushes less. A full node has full resources; it pushes the rated maximum. This is what "bandwidth limited" means at the source.

The receiver j imposes no additional limitation. Accepting incoming flux costs nothing at the receiver: the reservoir has empty space and the inflow simply fills it. There is no rate at which an empty reservoir "accepts" more slowly. The throttle is one-sided, source-only.

4.2 Throttle prescription as a choice

The single-throttle is one prescription within a family. Two alternatives:

No throttle. $F = \alpha(R_i - R_j)_+$. The rule of Paper I as written. Steady state: linear Poisson, $\delta \propto 1/r$ at leading order. No higher-order corrections in the static spherical regime. Equivalent to Newton.

Double throttle. $F = \alpha(R_i/R_0)(R_j/R_0)(R_i - R_j)_+$. Both source and receiver throttle the flux. Steady state: $\nabla^2(R^3) = 0$, giving $R/R_0 = (1 - B/r)^{1/3}$. Different functional form, not Schwarzschild. As we will see in Sec. 6, the double throttle has a pathological consequence: the cascade collapses near a strong sink because both endpoints freeze when their bandwidths drop.

Single throttle, source-side. Eq. (3). Steady state: $\nabla^2(R^2) = 0$, giving $R/R_0 = \sqrt{1 - \beta/r}$. The Schwarzschild form. Cascade propagates without collapse.

We adopt the single-throttle as physically motivated by the source-receiver asymmetry, but the rule of Paper I does not select it uniquely. The paper's result is conditional on this choice; we are explicit about that.

4.3 Continuum limit

In the limit of small gradients, the discrete flux (3) becomes a continuum vector field. With the gradient pointing in the direction of *decreasing* R, the flux flows downhill, and

$$\vec{F} \;=\; -\alpha \frac{R}{R_0} \vec{\nabla} R. \tag{4}$$

Conservation of reserve, away from sources, requires $\partial R/\partial t = -\vec{\nabla} \cdot \vec{F}$. In steady state this is

$$\vec{\nabla} \cdot \left[\frac{R}{R_0} \vec{\nabla} R \right] \;=\; 0. \tag{5}$$

5 Derivation of the Schwarzschild form

5.1 Reduction to Laplace equation for R^2

The key algebraic step. We use

$$\frac{R}{R_0} \vec{\nabla} R \;=\; \frac{1}{2R_0} \vec{\nabla}(R^2).$$

Substituting into Eq. (5):

$$\frac{1}{2R_0} \nabla^2(R^2) \;=\; 0 \quad \Longleftrightarrow \quad \boxed{\nabla^2(R^2) \;=\; 0.} \tag{6}$$

The throttle has converted the conservation law from "R is harmonic" (linear Poisson) into "R^2 is harmonic". The square is what carries the Schwarzschild structure.

5.2 Spherically symmetric solution

We seek a spherically symmetric solution of Eq. (6), with boundary condition $R \to R_0$ at $r \to \infty$. The general spherically symmetric harmonic function in three dimensions is $A + B/r$. Therefore

$$R^2(r) \;=\; R_0^2 \left(1 - \frac{\beta}{r}\right), \qquad R(r) \;=\; R_0 \sqrt{1 - \frac{\beta}{r}}, \tag{7}$$

for some constant β with units of length. The square root is real for $r \geq \beta$, with $R = 0$ at $r = \beta$; this is a horizon of the model, where the bandwidth vanishes.

5.3 Determination of β from the source

The constant β is fixed by the source strength. Compute the radial flux at radius r:

$$F_r(r) \;=\; -\,\alpha\,\frac{R}{R_0}\,\frac{\partial R}{\partial r}.$$

From Eq. (7), $\partial R/\partial r = R_0\,\beta/[2r^2\sqrt{1-\beta/r}]$, and the prefactor combines with the square root to give

$$\frac{R}{R_0}\,\frac{\partial R}{\partial r} \;=\; \frac{R_0\,\beta}{2r^2}.$$

Therefore $F_r(r) = -\alpha R_0\beta/(2r^2)$, independent of r. The minus sign indicates flow inward, toward the sink. The total flux entering a sphere of radius r is

$$\Phi \;=\; -\oint F_r\,dA \;=\; \frac{\alpha R_0\beta}{2r^2}\cdot 4\pi r^2 \;=\; 2\pi\,\alpha\,R_0\,\beta,$$

also independent of r, as required by stationarity.

This entering flux is absorbed by the sink at rate κE_0:

$$\kappa E_0 \;=\; 2\pi\,\alpha\,R_0\,\beta \quad\Longleftrightarrow\quad \boxed{\beta \;=\; \frac{\kappa E_0}{2\pi\,\alpha\,R_0}.} \tag{8}$$

5.4 Comparison with the linearised regime

Expand Eq. (7) for small β/r:

$$\frac{R(r)}{R_0} \;\approx\; 1 - \frac{\beta}{2r} - \frac{1}{8}\left(\frac{\beta}{r}\right)^2 - \cdots$$

The leading-order deficit is $\delta(r)/R_0 \approx \beta/(2r) = \kappa E_0/(4\pi\alpha R_0 r) = M_{\text{eff}}/(4\pi R_0 r)$, where $M_{\text{eff}} = \kappa E_0/\alpha$. This is exactly the Newtonian profile of Paper I. The throttle adds a quadratic correction; the linear regime is recovered as the leading term, and the higher-order structure is generated by the cascade.

5.5 Identification with Schwarzschild

Under the calibration of Paper I matching the dimensionless deficit to the physical Newtonian potential at leading order, $M_{\text{eff}}/R_0 \leftrightarrow 2GM_{\text{phys}}/c^2$, we have $\beta = 2GM_{\text{phys}}/c^2$. Substituting into Eq. (7):

$$\frac{R(r)}{R_0} \;=\; \sqrt{1 - \frac{2GM_{\text{phys}}}{c^2 r}}. \tag{9}$$

This is exactly the time-time component of the Schwarzschild metric in the weak-field limit, $\sqrt{|g_{00}|}$. Combined with the bandwidth identification (2):

$$\frac{d\tau(r)}{dt} \;=\; \sqrt{1 - \frac{2GM_{\text{phys}}}{c^2 r}}. \tag{10}$$

The match holds at all orders in $GM/c^2 r$, in the continuum limit, modulo the single-throttle prescription. The leading order coincides with weak-field general relativity by calibration; the higher orders are generated by the bandwidth-throttled cascade and coincide with the Schwarzschild expansion exactly.

6 Numerical illustration of the cascade

We implement the rule of Paper I together with three flux prescriptions: no throttle (recovering Paper I); single throttle, Eq. (3); and double throttle, $F = \alpha(R_i/R_0)(R_j/R_0)(R_i - R_j)_+$. We run each on a cubic substrate with a single sink at the centre and let the system reach steady state.

6.1 Setup

We use $L \in \{32, 40\}$, $\alpha = \kappa = R_0 = 1$, sink strength $E_0 \in \{0.5, 1.5\}$, time step $\tau \leq 0.05$, and run for several thousand ticks. Boundary: Dirichlet, with the outermost layer fixed at R_0. The script `code/iterative_cascade_test.py` executes all three prescriptions in turn and records the radial deficit profile.

6.2 Results

Table 1 reports the radial deficit profile from a representative run at $L = 40$, $E_0 = 1.5$. The "GR target" column is the continuum Schwarzschild prediction $1 - \sqrt{1 - 2x}$ with $x = M_{\text{eff}}/(4\pi r R_0)$.

r	x	linear	single throttle	double throttle	GR target
1.75	6.82×10^{-2}	5.99×10^{-2}	6.14×10^{-2}	$\sim 2 \times 10^{-6}$	7.07×10^{-2}
2.75	4.34×10^{-2}	3.66×10^{-2}	3.73×10^{-2}	$\sim 2 \times 10^{-6}$	4.44×10^{-2}
3.75	3.18×10^{-2}	2.68×10^{-2}	2.71×10^{-2}	$\sim 2 \times 10^{-6}$	3.24×10^{-2}
4.75	2.51×10^{-2}	2.00×10^{-2}	2.02×10^{-2}	$\sim 2 \times 10^{-6}$	2.55×10^{-2}

Table 1: Radial deficit profile $\delta(r)/R_0$ from the cascade simulation, $L = 40$, $E_0 = 1.5$. The single throttle adds a positive correction to the linear profile in the direction of the GR target. The double throttle collapses to near zero deficit beyond the immediate neighbourhood of the sink, illustrating cascade freezing.

6.3 Verifying the Schwarzschild form

The simulation underestimates the continuum deficit by a common lattice factor at $L = 40$ (the same finite-volume under-resolution observed in Paper I, Sec. 6.2). The clean test of the analytic derivation is therefore not single-vs-GR in absolute terms, but single-vs-Schwarzschild evaluated at the lattice's effective $x_{\text{eff}} = \delta_{\text{linear}}/R_0$.

r	$x_{\text{eff}} = \delta_{\text{lin}}/R_0$	$1 - \sqrt{1 - 2x_{\text{eff}}}$ (predicted)	single sim	discrepancy
1.75	0.05993	0.06184	0.06142	-0.7%
2.75	0.03665	0.03735	0.03725	-0.3%
3.75	0.02676	0.02713	0.02708	-0.2%
4.75	0.01998	0.02018	0.02016	-0.1%

Table 2: Test of the analytic Schwarzschild form against the single-throttle simulation. We compute the predicted $1 - \sqrt{1 - 2x_{\text{eff}}}$ using the lattice-measured deficit of the linear simulation as the effective x. The agreement with the single-throttle simulation is at the sub-percent level across the bulk fit window.

The agreement is sub-percent. The analytic derivation of Sec. 5 is verified by the simulation within the lattice resolution.

6.4 Cascade collapse under double throttle

The double throttle entries in Table 1 are essentially zero (modulo numerical noise) beyond the immediate vicinity of the sink. This is not numerical failure: it is the physical pathology of the double throttle. When R drops near the sink, both endpoints of the adjacent links carry low-bandwidth gates, the flux is suppressed quadratically, and the cascade freezes. The sink cannot drain its second-nearest neighbours efficiently because the link to the first-nearest neighbour — which has low bandwidth — is itself throttled. The result is an artificial horizon at the first shell.

The single throttle does not have this property: only the source endpoint is throttled, so a low-bandwidth source can still receive freely from a high-bandwidth neighbour, and the cascade propagates. The simulation thus distinguishes the two prescriptions on a structural basis, supporting the single-throttle as the physically natural choice.

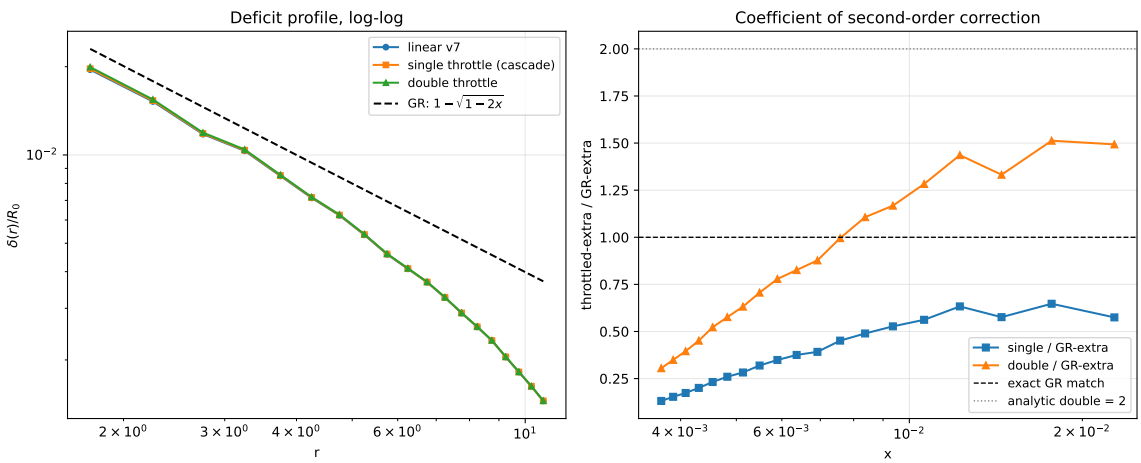

Figure 1: Cascade simulation. Left: log–log radial deficit profile for the linear, single-throttle, and double-throttle prescriptions, against the Schwarzschild target. The double throttle freezes near the sink. Right: the second-order correction added by each throttle, normalised to the expected GR correction (dashed: 1, exact GR match; dotted: 2, the analytic prediction for the double throttle in the continuum). The single throttle approaches unity at small x (within lattice resolution).

7 Calibration vs. prediction

We separate clearly:

Calibrations.

- Bandwidth identification (2): an operational hypothesis, not a derivation.
- Single-throttle prescription (3): a physically motivated choice (source-receiver asymmetry), not a derivation.
- $M_{\text{eff}}/R_0 \leftrightarrow 2GM_{\text{phys}}/c^2$: fixes one combination of lattice unit and rest level by demanding the leading-order deficit match the Newtonian potential. Inherited from Paper I.

Derived from these inputs.

- The full functional form $R/R_0 = \sqrt{1-2GM/c^2r}$, to all orders in GM/c^2r in the continuum limit, follows from the steady-state of the throttled rule, the spherical symmetry, and the source-strength match.

- The location of the bandwidth horizon, $r = 2GM/c^2$, is the locus where $R = 0$ in the analytic solution.
- The flux conservation law $\kappa E_0 = 2\pi\alpha R_0\beta$ relating source strength to the integrated flux across any sphere outside the sink.

8 What this paper does not do

We do not derive the Schwarzschild form for non-spherical or non-static configurations. The derivation in Sec. 5 uses spherical symmetry and steady state. Rotating, oscillating, or multi-source configurations are not covered.

We do not derive the spatial part of the Schwarzschild metric. The derivation gives the g_{00} factor only. The radial component, the angular components, the propagation of light through the modified bandwidth field, and any other component of the metric structure are not addressed.

We do not derive the Einstein field equations. The dynamical content of GR — the Bianchi identities, the matter coupling, the propagation of gravitational waves — is outside the scope of this paper.

We do not derive the single-throttle prescription itself. It is a physically motivated choice within a family of throttle prescriptions, justified by the source-receiver asymmetry. The rule of Paper I as written admits multiple throttle prescriptions; the result of this paper is conditional on the single-throttle choice.

We do not address kinematic time dilation. The motion of a localised configuration across the substrate, and any relationship to the special-relativistic Lorentz factor, is the subject of Paper III.

We do not derive Newton's G or the speed of light c. They enter as calibrations, exactly as in Paper I.

9 Relation to existing work

Weak-field general relativity. The match (10) reproduces the Schwarzschild time-dilation factor of weak-field GR at all orders in GM/c^2r. The match is calibration-mediated at leading order; the higher orders follow from the bandwidth-throttled cascade in the rule.

Emergent gravity programmes [1–3]. These derive gravitational behaviour from thermodynamic or information-theoretic arguments on horizons. The present paper operates at a different level: an explicit local rule with an explicit throttle prescription, rather than a coarse-grained thermodynamic identity. The two are not in competition; they target different levels of description.

Lattice and discrete models of gravity [4–6]. Lattice fluid models recover continuum hydrodynamic equations through similar steady-state and conservation arguments. The present paper differs in two respects: the bandwidth identification (2) maps the substrate's flux activity onto proper time, which is, to our knowledge, novel within the local-rule programme; and the single-throttle modification of the link rule is the specific structural ingredient that produces $\nabla^2(R^2) = 0$ rather than $\nabla^2 R = 0$, upgrading Newton to weak-field Schwarzschild.

10 Code and data availability

The Python script `code/iterative_cascade_test.py` implements the rule of Paper I with three flux prescriptions (none, single throttle, double throttle) and produces the radial profiles of Sec. 6. The script is pure-numpy and depends on no external simulation framework. The repository is available at https://github.com/stanislasdewavrin/DDD-Paper-2-Schwarzschild.

The numerical data of Tables 1 and 2 are stored in `data/iterative_cascade_L40.json`.

11 Open questions

Origin of the throttle prescription. The single-throttle is selected by physical motivation (source-receiver asymmetry of flux exchange) but not derived from the rule of Paper I as written. A microscopic argument that selects the single throttle, or that selects a specific asymmetry between source and receiver from a deeper property of the rule, would close the most important remaining gap.

Origin of the bandwidth identification. The identification of effective proper time with available bandwidth is operational. A microscopic argument that selects the linear dependence (2) from the rule, or that establishes proper time as a derived rather than an adopted quantity, would strengthen the construction.

Beyond the linear regime. When a node's reserve approaches zero (near the bandwidth horizon $r = \beta$), the rate-limiter β_i of Paper I becomes active and the rule departs from the linearised regime. The physical interpretation of the horizon, and the behaviour of the rule in its immediate vicinity, are not addressed in this paper.

Spatial part of the metric. The bandwidth identification gives g_{00}. Whether the substrate also produces a consistent spatial metric, and whether photon trajectories computed from the bandwidth field reproduce the deflection angle $4GM/c^2b$ of weak-field GR, is the subject of Paper IV.

Back-reaction. The probe approximation ignores the test configuration's contribution to the deficit field. The full self-consistent treatment is also left to Paper IV.

12 Outlook

1. Paper III — discrete-time response, inertial effects, and kinematic clock-rate suppression of moving configurations, with cosmogenesis from a single founding impulse as a corollary of the no-deadlock property of the rule. *Speculative.*

2. Paper IV — predictions and tests of the gravitational sector beyond the static spherically symmetric scenario: photon deflection, compact-source phenomenology near the bandwidth horizon, self-consistent feedback, and laboratory bounds. *Mixed.*

3. Paper V — the gauge sector and emergent electromagnetism. *Speculative; coefficients uncalibrated.*

4. Further exploratory notes (Papers VI and onward) address coupling, cosmology, quantum phenomenology, particle structure, and condensed-matter analogies. *Increasingly speculative.*

The present paper claims one structural result: under the single-throttle modification of the rule of Paper I and the bandwidth identification, the static spherically symmetric deficit field is the Schwarzschild time-dilation factor at all orders in the continuum limit. The result is conditional on the throttle prescription and on the bandwidth identification, both of which are made explicit. The rest is what one might do with such a rule.

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